SLCT — SURGEON LEARNING CURVE TRACKING

SLCT · v0.3

Study / Protocol	Site / Code	SLCT Record ID	Record Status
Surgeon Name / ID	Procedure Family / Anatomical Field	Device / System	Date Opened
Current SCR / CCDM / TRE Reference		Applicable CIP / Procedure / IFU Version	**Review type** ☐ Initial ☐ Early stabilization ☐ Periodic ☐ Triggered

A Purpose, Decision and Boundary

This log follows the procedural stabilization of one **surgeon** performing one defined **procedure family** with one defined **device/system**. It records context-sensitive signals close to the operative act so that surgeon-dependent variation can be reviewed before it appears as a later safety, deviation or data-quality problem.

Decision produced: Is execution within the defined scope becoming and remaining sufficiently stable for the current case-complexity context, or is intensified review, retraining, proctoring, restriction or formal escalation required?

This is not an entry credentialing record, a device-training record, an adverse-event form, a device-deficiency form or an intraoperative source record. It receives selected, traceable information from those records and documents the learning-curve interpretation and resulting control decision.

B Comparable-Case and Complexity Framework

Complete this framework when the log is activated, before case entry. It is the fixed comparison key for the surgeon, procedure family and device/system. Update it only when the procedure, device, indication, technique, complexity definition or applicable control changes. Each case record then refers to this framework rather than redefining the comparison in every line.

Comparison element	Local definition or rule to apply	Evidence source / document reference	Effective date / update
Procedure / device scope	State the procedure family, device/system, configuration range and any excluded approach or component.	**CIP:** **IFU:** **SCR:**	
Case-complexity strata	Define the lower, standard and higher-complexity contexts relevant to this investigation. Use only factors that materially affect interpretation of execution or duration.	**Procedure plan:** **Specialty criterion:** **Local / SRR criterion:**	
Operative-duration comparator	State the reference time or range for comparable cases. Do not compare materially different complexity classes.	**Comparator source:** **Version / reference:**	
Expected initial support	State whether proctoring, co-surgery, direct observation or senior support is expected during early cases and under what conditions.	**SCR condition:** **Proctoring plan:** **CCDM level:**	
Review triggers	State repeated or material patterns that require review. Do not insert a numerical threshold unless justified for this study.	**CIP / SRR:** **Oversight / safety plan:**	

Enter the supporting document and its version, section or record ID. The source is a controlled document or record, not the name of a reviewer.

C Case-Level Tracking — Initial Cases 1 to 5

Case seq.	Date / case source reference	Procedure / device / configuration	Complexity class and key factor	Operative duration / comparator	Unplanned substitution or technical deviation	Senior support / proctoring	Documentation / early technical outcome / linked record
1							
2							
3							
4							
5							

Record a concise source reference, not participant-identifying information. State “none observed” where an expected variation, deviation or support event did not occur; do not leave a negative finding ambiguous.

D Initial-Cohort Review

Review the first cohort against the comparison framework. Interpret patterns across comparable cases rather than any isolated event, and record a linked action whenever a concern or material limitation is identified.

Review question	Evidence and context assessed	Conclusion	Required action / linked record
Are operative duration and support needs interpretable within comparable complexity strata?		☐ Stable ☐ Observation ☐ Concern	
Are substitutions or technical deviations isolated, expected or showing recurrence?		☐ Isolated / expected ☐ Observation ☐ Concern	
Do documentation, imaging or early technical findings suggest a procedural pattern?		☐ No pattern ☐ Observation ☐ Concern	

E Scheduled Review Register

Use one row for each planned cohort or periodic review. Summarize the complexity range, the observed pattern, the decision made and the next review or linked control record.

Review point / period	Cases and complexity context reviewed	Pattern across duration, variation, support and evidence quality	Decision / next review / linked records
Cases 1–5			
Cases 6–10			
Periodic review			
Other defined cohort			

F Case-Level Tracking — Continuation

Case seq.	Date / case source reference	Procedure / device / configuration	Complexity class and key factor	Operative duration / comparator	Unplanned substitution or technical deviation	Senior support / proctoring	Documentation / early technical outcome / linked record
6							
7							
8							
9							
10							

Attach continuation pages using the same column order for additional cases. Keep a distinct log for a materially different procedure family or device/system unless the approved comparison framework explicitly permits aggregation.

G Triggered Learning-Curve Assessment

Open this assessment when a pre-defined or material signal appears. Triggers may include repeated unplanned component substitutions, duration consistently outside the comparable-case expectation, recurrent technical deviations, clustering of early technical complications, repeated need for senior support, a new device/technique/indication, or an external signal from CSST, SRR, IMC or PI oversight.

Trigger and date	Cases / records reviewed	Complexity-aware assessment	Likely control issue, if any	Immediate disposition

Do not attribute a safety signal to surgeon performance without reviewing complexity, device, procedure, patient and documentation context. A confirmed complication signal should be assessed against this log before it is attributed to the device or to the operator.

H Action, Restriction and Reassessment

Signal / assessment reference	Action or restriction	Owner / due date	Evidence required for reassessment	Outcome / linked record / closure

A restriction, training intervention or intensified review is not closed merely because it was initiated. Record the reassessment evidence and whether the control was effective, insufficient or still under review.

I Disposition Logic

Use this logic only after completing the scheduled review or a triggered assessment. It converts the conclusion into a controlled response. The selected outcome is recorded in the action log and in the overall status below; it is not a second assessment or a substitute for the trigger table.

Disposition	Meaning	Required destination or response
Continue as planned	The available evidence is stable within the approved scope and comparable case-complexity context. No material surgeon-dependent drift is identified.	Record next scheduled review and retain source references.
Intensify review	A pattern is uncertain, early or limited. It does not yet justify a scope restriction but requires closer observation or a shorter review interval.	Define additional cases, reviewer and next review date.
Adapt controls	Evidence supports a local intervention such as renewed device training, temporary proctoring, co-surgery, enhanced documentation review or restriction to lower-complexity cases.	Update TRE, SCR or CCDM as applicable; open a dated action and reassessment point.
Formal escalation	A material, recurrent or unresolved pattern requires risk, governance, safety or sponsor review according to the applicable procedure.	Open or update SRR / CAPA; notify PI oversight, CSST or sponsor route as applicable.

J Overall Record Status and Certification

I confirm that this log was reviewed against the applicable source records for the surgeon, procedure family and device/system stated in the header. The conclusion reflects complexity-aware interpretation of the recorded data. It does not replace safety assessment, device deficiency assessment, participant-level clinical judgment or the PI's retained responsibility for the clinical investigation.

Current learning-curve status	Select one and state the current scope or condition	Effective date / next review
Stable / continue	☐ Current procedural scope and supervision remain appropriate.	
Under intensified review	☐ Additional cases or a shorter review interval are required.	
Control adaptation active	☐ Training, proctoring, scope or documentation control has been changed.	
Escalated / restricted	☐ Formal risk, governance, safety or sponsor route has been activated.	

Primary reviewer

Name / role:

Signature:
Date:

Surgical / clinical reviewer

Name / role:

Signature:
Date:

Principal Investigator / accountable approver

Name:

Signature:
Date:

Next scheduled review / controlled record location

Review date:

Record location:
Current version: